Build an Atom

Three particles, three separate jobs — and where the decimal under chlorine actually comes from.

Name ____________________ Date ____________________ Period ____________________

Demonstration: andresdbp.github.io/resources/demos/build-an-atom/

1 Before you open anything

Answer from what you already know, on paper, before the demonstration is on screen. The point is to have a commitment to compare against later.

PREDICTION

The periodic table prints **35.45** under chlorine. One chlorine atom is picked at random out of an ordinary sample of table salt. Write down the *mass number A* you expect that single atom to have, and the *mass in u* you expect it to have.

Now say in one sentence where you think the number 35.45 came from.

2 Three particles, three jobs

Open the demonstration. The **Particles** panel has three sliders and a pair of buttons under each. The **Identity** panel above reports what you have built.

1. In **Try these**, click **carbon-12**. Fill in row 1.
2. Click **+ neutron** twice. Fill in row 2.
3. Click **chlorine-35**, then click **+ electron** once. Fill in row 3.
4. Click **sodium Na^+**. Fill in row 4.
5. Now set the three sliders by hand: **Protons 19**, **Neutrons 20**, **Electrons 18**. Fill in row 5.

p, n, e	Element name	Z	A	Net charge	Valence electrons	Chip in the Identity card (neutral atom / cation / anion)
6, 6, 6						
6, 8, 6						
17, 18, 18						
11, 12, 10						
19, 20, 18						

a. Between rows 1 and 2 you added two neutrons. List every readout in the table above that changed, and every one that did not.

b. In row 3 you added one electron to chlorine-35. Which readouts changed? Is the result still chlorine? How do you know from the screen?

c. Complete each sentence with *protons*, *neutrons* or *electrons*.

The _____________ decide which element it is. The _____________ decide which isotope it is. The _____________ decide the charge.

d. Write the full nuclide notation shown in the Identity panel for row 5, including the charge.

3 Where 35.45 comes from

Click **chlorine-35** in **Try these**, then scroll to the panel titled **Isotopes — where 35.45 comes from**. Under **Mix the abundances** there is one slider per isotope; moving one makes the others share out whatever is left, so the two always total 100%.

1. Drag the **Cl-35** slider all the way to the right (100%). Record the **Weighted average atomic mass, your mix** readout.
2. Click **Back to natural abundances** and record both numbers again.
3. Drag **Cl-35** as close to 50% as you can, then as close to 25% as you can. Each time, write down the percentage the readout actually shows, not the one you aimed for.
4. Drag **Cl-35** all the way to the left (0%) and record.

What you set	Cl-35 % shown	Cl-37 % shown	Weighted average atomic mass, your mix (u)
Cl-35 at maximum			
natural abundances			
Cl-35 near 50%			
Cl-35 near 25%			
Cl-35 at minimum			

a. Two of your five averages are also the mass of a single real atom. Which two rows, and what makes those rows different from the other three?

b. State the rule in your own words: what happens to the weighted average as the Cl-35 percentage falls, and between which two values is the average trapped no matter what you set?

c. With the natural abundances restored, the average is much nearer 35 than 37. Give the reason, in terms of the abundance column.

d. Go back to your prediction in Part 1. Which parts of it does the demonstration confirm, and which parts does it contradict? Be specific about the number you wrote for the mass of a single atom.

> The masses in the **Mass (u)** column are measured values in unified atomic mass units. Only carbon-12 reads exactly 12.000000, because 1 u is *defined* as one twelfth of a carbon-12 atom. Every other nuclide lands a little off its whole number.

4 Working backwards, and a shortcut that fails

Exam questions rarely hand you the abundances and ask for the average. They usually run the other way.

a. Read the two isotope masses out of the **Mass (u)** column and copy them here, to four decimal places.

Cl-35 = _____________ u Cl-37 = _____________ u

b. Let *f* be the fraction of chlorine atoms that are Cl-35, so the fraction that are Cl-37 is (1 - *f*). Set up the equation that makes the weighted average come out to 35.453 u, and solve it for *f*. Show the algebra.

c. Click **Back to natural abundances** and compare your *f* with the Cl-35 abundance the table lists. Agree to how many decimal places?

d. A shortcut worth testing. A quick way to get an average of two isotopes is to average their masses: (34.9689 + 36.9659) ÷ 2. Compute it, then drag the Cl-35 slider as near 50% as you can and read the average off the screen.

Under what condition does that shortcut give the right answer, and why does it fail for chlorine?

e. A second shortcut: use the mass *numbers* 35 and 37 instead of the measured masses. Compute 35(0.7576) + 37(0.2424) and compare it with the 35.453 u on screen. Is the error large enough to matter at three significant figures? Explain what causes it.

5 Solve these without the demonstration

Close the page or look away. Show all working. Then reopen it, set the element with the **Protons** slider, and check yourself — if a number is off, write down where.

1. Boron has two natural isotopes: B-10 at 10.0129 u, 19.9% abundant, and B-11 at 11.0093 u, 80.1% abundant. Calculate the average atomic mass to two decimal places. Then state, without calculating, whether your answer should be nearer 10 or nearer 11, and why.

2. Lithium has exactly two natural isotopes, Li-6 at 6.0151 u and Li-7 at 7.0160 u. The periodic table value is 6.94 u. Find the percentage abundance of each isotope.

3. An ion contains 16 protons, 18 neutrons and 18 electrons. Give its element name, its atomic number Z, its mass number A, its net charge, and write it in nuclide notation with the charge included.

4. Potassium-39 has a mass of 38.9637 u and makes up 93.26% of natural potassium, yet the periodic table prints 39.098 u for potassium. Explain, without doing the full calculation, why the printed value is larger than 38.9637 — and explain why no potassium atom you could ever isolate has a mass of 39.098 u.

6 On your own

Nothing here is graded, and the demonstration does more than this worksheet used.

- Drag **Protons** to 0 and read what the Identity panel and the isotope chart say. Then rebuild an atom one proton at a time and watch where the element name first appears.
- Find an element in the demonstration that has only one natural isotope — beryllium, fluorine, sodium, aluminium and phosphorus all qualify. Its periodic table value is still not a whole number. Read the note the panel gives and work out why.
- Build a nuclide that nature does not stock in weighable amounts, such as **carbon-14**, and read the caption under the atom. Then check the isotope chart: where is its bar, and what does that tell you about its contribution to the average?
- Set **Protons** to 20 and count the shells. The **Honest limits** panel explains why the demonstration stops there. Predict what the shell diagram would get wrong for element 21.